
IRQG: A Multi-Agent Framework for Educational Question Generation

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Abstract

Generating high quality Multiple Choice Questions (MCQ) in specialized domains from source material remains extremely challenging. Single pass MCQ generation comes with few guarantees of quality, and accurate information retrieval becomes inaccurate as source material length grows. We propose IRQG,¹ a multi-agent refinement pipeline that consists of: (1) a retrieval paradigm to identify most relevant source material sections, (2) a question generator agent (3) a test-taking agent, (4) a feedback agent, and (5) a refinement agent tasked with improving the original generated questions. In a human study with questions aimed to align to course learning objectives, refined items are preferred in 73.6% of head-to-head comparisons, while relevance to learning objectives rises from 42% to 85% on average. Results indicate that decomposition into generation, test-taking, feedback, and refinement is a practical recipe for improving MCQ quality.

1 Introduction

Question Generation (QG) is a core NLP task with applications across many industries [1; 2]. Ensuring that QG systems produce high quality questions is integral to creating effective assessments and facilitating adaptive learning systems. Even though Vision Language Models (VLMs) have significantly improved the QG process from multimodal source material (e.g., charts, graphs, tables) in recent years [3], ensuring that the questions are high quality is still very difficult. Current QG systems struggle with context selection and question quality especially when the context is very long [4]. This is an issue particularly prevalent in multimodal contexts, because it is difficult to identify which related information is also useful and salient for QG. Dense bi-encoder retrieval methods allowed for scalable and efficient context selection, outperforming sparse methods for open source QA [5]. Late-Interaction retrieval was introduced [6] showing that fine-grained matching boosted retrieval accuracy over bi-encoder methods, and ColPali adapted ColBERT for VLM retrieval achieving state of the art results in document-based search [7].

Single-pass, single-agent QG often entangles all the skills required into one prompt, leading to inconsistent behavior that relies very heavily on the prompt. We hypothesize that a multi-agent setup would be able to decompose the task into smaller chunks, enabling critique-and-revision, cross-checking for a single correct answer, and better alignment with learning objectives. Multi-stage refinement pipelines improve grammar and relevance of questions over single pass generation [8], and critique-and-revision prompting improved question/answer clarity and correctness in multiple choice questions [9]. We propose an Iterative Refinement for Question Generation (IRQG) pipeline that combines vision-language retrieval, question analysis, and question refinement that improves QG quality.

¹IRQG Stands for "Iterative Refinement for Question Generation".

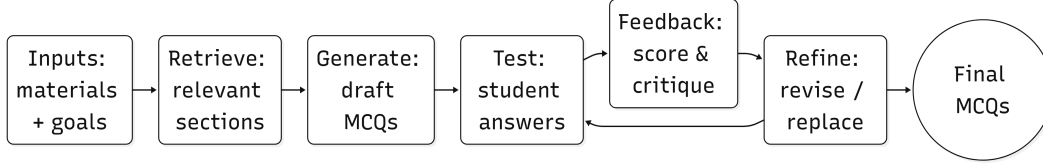


Figure 1: High-level overview of the IRQG pipeline.

2 Architecture

At a high level, as shown in Figure 1, our goal is to create a system that produces high-quality MCQs q (with associated answer rubrics), given a textbook b and a set of learning objectives $L = \{l_1, \dots, l_n\}$. We find that the incorporation of a learning objective in the process is paramount for the creation of meaningful questions.

2.1 The IRQG Pipeline

Our pipeline aims to follow how humans would determine question quality. In traditional assessment design, educators first create a test blueprint, aligning content areas and cognitive demand, followed by iterative revisions informed by student performance and misconceptions [?]. Our agent roles echo prior agentic designs for MCQ construction [10], but differ in objective (refinement of MCQ questions over conversion from open ended questions to MCQ) and supervision (human-grounded scoring). Concretely, our approach uses the following agents:

- **Retrieval Agent:** Receives a corpus of multi-modal source material b along with a learning objective l_i and retrieves the most relevant sections as a set of pages $P = \{p_1, \dots, p_m\}$. This will form the necessary more specialized context for the rest of the question generation pipeline to generate targeted questions rather than generic ones from a large context.
- **Question Generation Agent:** Given the textbook context P as well as the same learning objective l_i , it generates a set of preliminary questions $Q^0 = \{q_1, \dots, q_k\}$ along with corresponding answers $A^0 = \{a_1, \dots, a_k\}$.
- **Test-Taking Agent:** Receives the questions Q^0 as a test, without having access to the source material, producing answers T^0 . The purpose of this agent is to simulate a human taking a test. This helps determine if questions and answer choices are confusing for someone without access to the source material.
- **Feedback Agent:** Receives the original questions Q^0 with their answers A^0 and the answers T^0 of the test-taking agent. Generates an overall score and feedback f_i for each question q_i based on a set of pre-defined evaluation questions that capture different dimensions of MCQ quality: whether the question is grammatical and clear, the answer choices clear, if there is only one correct answer choice. This simulates an instructor that receives a student's answers to a test, aiming to improve their test design.
- **Refinement Agent:** Last, this agent receives the original questions Q^0 , the feedback $F^0 = \{f_1, \dots, f_k\}$ from the feedback agent, and the same source material P and l , and generates a set of updated questions (if needed) Q^1 with associated answers A^1 .

This refinement loop $(Q^0, A^0) \rightarrow T^0 \rightarrow F^0 \rightarrow (Q^1, A^1)$ could in theory keep iterating to produce further updated questions (Q^2, A^2) and so on, but in practice we find that a single refinement iteration is enough, with any additional iterations introducing minimal changes.

2.2 Experimental Setup

The source material b is initially indexed using Colpali [7]. Then a learning objective l_i is used as a query to search through b using late interaction retrieval [6], which computes the relevance score between the query Q and a document D as:

$$\text{score}(Q, D) = \sum_{q \in Q} \max_{d \in D} \mathbf{q}^\top \mathbf{d}, \quad (1)$$

where Q, D are the sets of contextualized token embeddings for the query and document, $\mathbf{q}, \mathbf{d}$ are token vectors from a shared encoder, $\mathbf{q}^\top \mathbf{d}$ is the dot product, and $\max_{d \in D}$ selects, for each q , the most similar d .

We implement the Question Generation Agent using a VLM to produce initial questions (Q^0, A^0) from the retrieved pages P , to ensure any non-textual elements (figures, graphs) are also potentially taken into account.

For the refinement pipeline (§2.1), we use the following models for each component:

- **Initial Question Generation Agent:** Qwen-2.5-VL-7B [11]
- **Test-Taking Agent:** Claude Sonnet 3.7
- **Feedback Agent:** Claude Sonnet 3.7
- **Refinement Agent:** Claude Sonnet 3.7

3 Experiments and Results

We conducted a study on a specialized domain (NLP) to evaluate the proposed refinement approach. We generated 25 original/refined question pairs from an advanced NLP textbook, tied to learning objectives taken from the course’s syllabus.²

We then tasked 5 NLP experts (senior PhD students who have TAed for a graduate NLP course in the past) with providing feedback using the same criteria that the feedback agent did.

Does refinement improve QG? The original and refined questions were presented side by side in an online annotation interface built within LabelStudio [12]. Our first question to the evaluators was which of the two questions (presented in random order) was better.

Overall, we find strong indicators that the refined questions were significantly better than the original ones. Table 1 shows the rate in which different graders picked certain choices when presented with an original/refined question pair.

	Preferred Refined	Preferred Original	Tie
All graders	92 (73.6%)	10 (8.0%)	23 (18.4%)
Grader 1	19	2	4
Grader 2	17	0	8
Grader 3	20	2	3
Grader 4	19	4	2
Grader 5	17	2	6

Table 1: Preference counts across 125 judgments (25 items $\times$ 5 graders). Refined questions are preferred more than 70% of the time.

Are the Questions Good? An important consideration is whether the produced questions are *relevant* and appropriate with respect to the learning objective. We explicitly asked our NLP experts to judge whether the produced questions were relevant to the associated learning objective. Results are shown in Appendix Table 4. All graders are in agreement that refined questions are relevant to the learning objective about 85% of the time, in contrast to only about 40% of the time for the original questions.

Beyond the head-to-head comparison, for each of the 25 original–refined question pairs we collected two kinds of signals: (i) an overall score and four binary criteria (SQ1–SQ4) that match the questions the feedback agent has to address. The averaged expert scores for each sub-question are listed in Table 5.

The subquestions are the same as the evaluation criteria by the feedback agent: (1) Is the question clear? (2) Are the answer choices clear? (3) Is there only one correct answer? (4) Is the question grammatically correct?

The main takeaways are:

²Anonymized for review.

- The produced questions do not suffer from grammar errors (SQ4);
- The questions and choices are generally clear (85+% originally), but the refinement process addresses these issues (98% addressed after refinement);
- about 30% of the original questions do not have a single correct answer in the produced choices. The refinement process largely addresses this issue as well (95% after refinement).

Is the Feedback Agent Good? The last element of our evaluation compares the human scores and answers with the ones produced by our Feedback Agent. We note again that we asked the human annotators the exact same questions we task the Feedback Agent with answering. First, in Table 2 we compare the overall scores: we note that the feedback agent and our human graders agree on the general quality of the original questions (with scores around 4/10). They also agree that the refined questions are better. However, human graders rate the refined ones at about 7/10 in terms of overall quality, while the feedback agent overestimates their quality (to 9/10).

Feedback Agent (Grading Model)		Human Graders	
Original	Refined	Original	Refined
4.04 ± 1.28	9.11 ± 0.78	4.61 ± 2.22	6.9 ± 1.6

Table 2: Overall Question Score comparisons for the feedback agent. While the agent overestimates the quality of the refined questions, it nevertheless correlates with human judgments.

Next, we compare our feedback agent’s responses to the four subquestions to the answers of the human graders, with results shown in Table 6. Since these are binary questions, we use a majority vote to determine the human label. The grading model’s accuracy is generally high, especially for SQ1, SQ3 and SQ4. However, for SQ2, the feedback agent only agrees with the humans only about 50% of the time. We speculate that this is because whether the choices are clear is ambiguous. We also note that SQ3, the deterministic version of SQ2, received a notable increase from both the feedback model and the human graders.

Additional Ablations & Analysis We compare multiple versions of IRQG to ensure that the questions are improving solely from the refinement pipeline and not just the quality of the model outputting the second set of questions.

To eliminate the possibility that the questions only improved because the quality of the Claude models, we evaluated 25 questions generated by Claude 3.7 with just an initial prompt. Those generated questions were then passed through the same grading model for evaluation. The average score was 8.44 ± 1.44 indicating that the refinement process referenced in 2 still worked better.

Then to eliminate that the question improvement was simply because Qwen 2.5 VL 7B [11] was too small of a model and thus not good at generating questions, the same pipeline mentioned in Section 2.1 was replicated using bigger entirely open source models (Qwen2.5-VL-32B-AWQ). Even though the pipeline was fully replicated with open source models (including test taker and feedback), to ensure that evaluation was fair, the initial questions and refined questions were also passed through Claude to ensure the Feedback Agent produces a comparable score. The results are shown in Table 3 and indicate that the pipeline still improves the scores.

Original Questions	Refined Questions
Original	Refined
7.73 ± 2.24	9.04 ± 0.94

Table 3: Evaluated score for full Qwen2.5-VL-32B-AWQ pipeline

4 Conclusion

In this work, we showed that this multi-agent approach improves QG over zero-shot LLM generation for questions generated from source material. Future work will include extending and evaluating the pipeline for distinct types of data (e.g charts, tables, graphs). It will also include training a model to try and replicate the entire refinement pipeline into reasoning steps within a smaller model (e.g 7B, 32B).

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A Agent Details

Below are the system prompts for each agent.

- **Test-Taking Agent:**

Test-Taking Agent System Prompt

You are an NLP student. Take this quiz. Just output the question number and your answers without any explanations.

[quiz]

- **Feedback Agent:**

Feedback Agent System Prompt

You are a QA grader for question generation. Evaluate the question along the following criteria, also taking the student's answer into consideration.

Criteria: [criteria]

Question: [question]

Student's Answer: [answer]

Correct Answer: [correct answer]

Output an array of JSON objects like this:

```
{
  "well_formatted": true,
  "criteria": {
    "has_clear_question": true,
    "has_one_correct_answer": true,
    "has_clear_answer_choices": true,
    "grammatically_correct": true
  },
  "score": 9.2,
  "explanation": "The question is clearly phrased..."
}
```

- **Refinement Agent:**

Refinement Agent System Prompt

Here is some feedback on the questions ...
{feedback}

ANY QUESTION WITH A SCORE LOWER THAN 8 ...
Any question with a score of 9 or higher
can stay the same.
8 to 9 can be tweaked. < 8 → delete & create new.

Learning objective: {learning_test}

Current questions: {curr_ques}

Make sure you output all questions...

B Results on Learning Objective Relevance

Table 4 shows the graders’ scores w.r.t relevance to the learning objective:

Grader	Original (%)	Refined (%)
Grader 1	56	84
Grader 2	36	84
Grader 3	36	88
Grader 4	52	76
Grader 5	28	92
average	42 ± 10.1	85 ± 7.4

Table 4: Percentage of questions judged as relevant to the learning objective. The refinement process certainly improves relevance.

C Extended Results Tables

Average expert score for each subquestion is shown in Table 5.

Subquestion	Original	Refined	Δ
SQ1 – Question clear	86.4 ± 7.4	98.4 ± 3.2	+12.0
SQ2 – Choices clear	88.0 ± 7.4	98.4 ± 2.0	+10.4
SQ3 – One correct answer	72.0 ± 12.0	95.2 ± 4.7	+23.2
SQ4 – Grammar correct	98.4 ± 2.0	100.0 ± 0.0	+1.6

Table 5: Fine-grained evaluation of the questions (showing percent (%) true per subquestion).

Feedback agent results for subquestions are shown in Table 6

Subquestion	Before	After	Δ
SQ1 – Question clear	90.9	81.8	−9.1
SQ2 – Choices clear	54.5	45.5	−9.0
SQ3 – One correct answer	72.7	86.4	+13.7
SQ4 – Grammar correct	100.0	100.0	0

Table 6: Accuracy (%) of grading model vs. majority human vote for the 4 subquestions, before and after refinement.